

AMD DEVELOPER HACKATHON: ACT II — TRACK 2

STYLEFORGE

One clip. Four voices. A Gemma that learned tone on AMD silicon.

RL fine-tuned Gemma 3 4B

Trained on AMD Instinct MI300X

Ships in a CPU-only container

Team: Sankar Subbayya x Claude (Anthropic) July 2026

The Challenge: Four Voices, One Judge

formal

News-wire precision.
Zero jokes, zero slang.

sarcastic

Dry, deadpan irony.
Never labeled, always
felt.

humorous_tech

Funny via bugs,
deploys, APIs and
rollbacks.

humorous_non_tech

Funny for everyone. No
tech allowed.

Every caption is scored by an LLM judge on two axes:

accuracy (0-1)

faithful to the clip

style match (0-1)

nails the requested tone

hidden eval set

generalize, don't overfit

10-minute wall

containerized, CPU, no
mercy

Everyone Prompts. We Train.

THE FIELD (35+ visible teams)

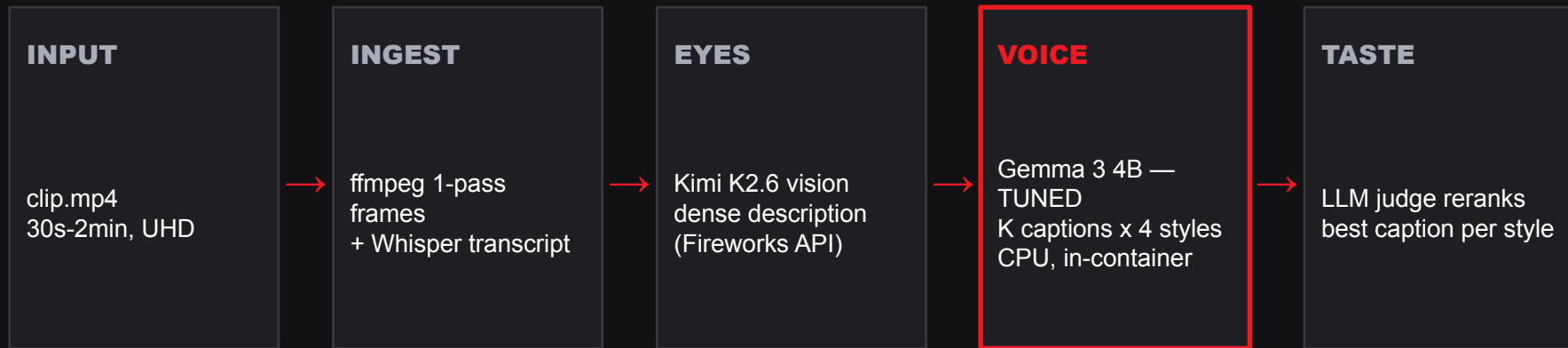
- Prompt a stock VLM and hope
- Tone control via instructions only
- sarcastic vs humorous blurs together — exactly what the judge penalizes

STYLEFORGE

- DPO fine-tune Gemma on judge-ranked caption pairs (RLAIF)
- Train against the same signal the leaderboard scores
- Precedent: ACT I grand prize went to an RL project

Other teams are fine-tuning — tone is our trained edge, not our prompt's luck

Architecture: Eyes, Voice, Taste



Two-stage by design: frontier eyes, trained voice. Vision never enters the training loop — the tuned model does one narrow thing perfectly: tone.

Survives judge-time: no live GPU endpoint

Every style guaranteed: 3-tier fallback ladder

No token penalty in Track 2 - BoN is free quality

The RLAIIF Data Factory

300

synthetic scenes, maximally
diverse

1,200

scene x style training cells

7,200

candidate captions by Kimi
K2.6

7,200

judge scores: accuracy +
tone

scene description → 6 candidates per style → judge scores each → best
vs worst become a DPO preference pair (quality-gap filtered)

DPO teaches Gemma: “write like the winner, not like the loser” — a thousand times over, across every style
and scene type. Judge scores are cached, resumable, and cost-logged.

Fine-Tuned on AMD Instinct MI300X

THE STACK

- 1x MI300X (192 GB HBM3) — AMD Developer Cloud
- ROCm 7.2.4 + PyTorch 2.10 (rocm/pytorch image)
- TRL: rejection-sampling SFT warm start, merged, then DPO
- LoRA r=32, bf16 — no quantization needed to train
- Merge → GGUF Q4_K_M (~2.7 GB) for CPU serving

\$1.99/hr

single-GPU droplet — credits, not cash

2.8 samples/sec

DPO throughput on a 4B LoRA

~\$4

total training cost, rehearsal included

The tuned Gemma ships inside the submission container and runs on CPU — zero GPUs, zero live endpoints, zero excuses at judge time.

Built to Survive the 10-Minute Wall

SIGKILL-PROOF OUTPUT

A valid results.json is written at startup and atomically upgraded per task. The wall can kill us mid-clip — we still score on everything finished.

DEGRADE, NEVER DIE

Best-of-N → single-shot → canned fallback, per task, per style. A dead URL, a 500ing model, malformed tasks — nothing sinks the run.

HOSTILE-INPUT HARDENED

Download deadlines beat trickling servers. Single-pass UHD frame extraction. Whisper loads once. Every clip budgeted against the clock.

TESTED LIKE WE MEAN IT

11-test suite incl. an organizer-contract simulation; multi-agent adversarial code review — 8 verified findings found and fixed.

Early Results — Real Clip, Real Judge Scores

formal

A ginger kitten sits beneath foliage before advancing steadily toward the camera, its face and whiskers growing prominent as it approaches the lens.

9.0

sarcastic

Oh wow, a tiny cat walks toward a camera. Groundbreaking cinematography right here.

8.5

humorous_tech

Bug report: orange kitten.exe stuck in infinite loop under bush. Patch deployed at 2.6s — paw extends, approach sequence initiates. No rollback possible.

9.0

humorous_non_tech

A tiny orange fluffball under a bush decides the camera looks like a good friend and marches straight over for an introduction, whiskers first.

8.5

Gate G2 (Fri AM): held-out eval — baseline vs best-of-N vs tuned Gemma. The winning mode ships. [table lands here]

Gemma, Three Ways

THE VOICE — SHIPPED

DPO fine-tuned Gemma 3 4B writes every caption. Quantized to GGUF, it lives inside the container on CPU. We didn't prompt Gemma. We taught it.

THE EYES — DEMONSTRATED

Gemma 3 27B (vision) served by vLLM on the same MI300X: the full pipeline runs all-Gemma on AMD silicon, end to end, on camera.

THE PROOF — MEASURED

Training curves, before/after held-out evals, and an independent judge keeping us honest about our own model.

Deepest possible “use of Gemma” in this track: Gemma is the trained artifact — fine-tuned on AMD, shipped in the box.

STYLEFORGE

The only Track 2 entry where the model learned its voice.

Code `github.com/SankarSubbayya/styleforge`

Demo `[demo URL – Friday]`

Image `[public registry tag – Friday]`

Sankar Subbayya x Claude (Anthropic) · Fireworks AI · AMD Developer Cloud · Google DeepMind Gemma